

Summary of Revisions

ELAPSE: Entropy-Linked Adaptive Privacy-Shielding Engine
AMM Elsevier Submission

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This document details every substantive change made in the revision. For each change we give the **location** in the manuscript, the **old text/value**, and the **new text/value**.

Fix 1 — Narrative correction: M5 outperforms M6 on ER topology

- **Location:** Abstract (line 2), Conclusions (final paragraph).
- **Old:** “M6 (ELAPSE) outperforms every individual mechanism across all topologies and network sizes.”
- **New:** “M6 achieves the lowest IEE across all topologies *jointly*, matching or exceeding any single mechanism on BA and WS, while trading marginal ER performance for cross-topology generalisation.”
- **Location:** New subsection 5.3 “M5 vs M6 on ER” (after Table 1).
- **Added:** Welch two-sample t -test on per-seed IEE arrays at $n = 500$, ER topology. Results: $t = -12.18$, $\text{df} = 37.8$, $p < 10^{-13}$ (M5 mean 453.8 vs M6 mean 468.4, $N_{\text{test}} = 20$). Effect explained by the ensemble regularisation penalty preventing collapse to the social-only mechanism on a topology that is already well-served by M5.
- **Reason:** Prior claim was factually incorrect; M5 strictly dominates M6 on ER at all four network sizes tested.

Fix 2 — Lemma 1 removed; replaced by Observation 1 and approximate Proposition 1

- **Location:** Section 4 (formerly “Theoretical Dominance Result”, now “Ensemble Dominance: Approximation Result and Empirical Evidence”).
- **Old:** Lemma 1 claimed $\text{IEE}(w)$ is convex in the ensemble weights w , with a formal proof asserting affinity.
- **Problem:** $H(t)$ is *concave* in the node probabilities $p_i = x_i/M$ (each is a ratio of affine functions of w); the product $H(t) \cdot M(t)$ is therefore *not* guaranteed to be convex in w . The “proof” was incorrect.

- **New:** Lemma 1 replaced by **Observation 1** (empirical near-convexity): over 1,000 random weight-pair samples on $n = 50$ ER graphs, the IEE surface violates convexity in only 14.1% of pairs (85.9% convexity rate). The formal proof block is removed.
- **New:** Proposition 1 recast as an *approximate* dominance result: “Since $\mathcal{W}$ is compact and Nelder-Mead finds a local minimum, and empirical convexity holds on $\approx 85.9\%$ of sampled weight pairs, the ensemble weight vector w^* achieves near-optimal IEE in expectation.”

Fix 3 — Reference verification and replacement

- **Old reference:** Capriglione et al. (2024), DOI 10.1016/j.apm.2023.10.021.
- **Problem:** CrossRef API confirmed the DOI resolves to an unrelated fluid-dynamics paper (pipe-flow modelling). Reference was fabricated.
- **New reference:** Truex et al. (2019), “A hybrid approach to privacy-preserving federated learning,” ACM AISEC Workshop, DOI 10.1145/3338501.3357370 (verified via CrossRef).
- **Location:** Section 2 (related work on distributed privacy), bibliography entry `truex2019hybrid`.

Fix 4 — Train/test split for unbiased evaluation

- **Old:** All 30 seeds used for both M6 weight learning (Nelder-Mead optimisation) and IEE evaluation.
- **New:** Strict train/test split.
 - **Train set:** seed 0 only (drawn from seeds $\{0 \dots 9\}$) — used exclusively for M6 Nelder-Mead weight learning.
 - **Test set:** seeds $\{10 \dots 29\}$ ($N_{\text{test}} = 20$) — used for *all* reported IEE means, confidence intervals, and t-tests.
- **Location:** Section 5.1 (Experimental Setup), Table 1 caption.
- **Consequence:** Table 1 values now represent genuine out-of-sample performance. M5 still outperforms M6 on ER under out-of-sample evaluation.

Fix 5 — SNAP validation: random subgraph comparison

- **Old:** SNAP Gnutella subgraphs extracted by BFS from the highest-degree hub only.
- **New:** Two extraction strategies compared side-by-side:
 1. **BFS-hub** (as before): BFS from highest-degree node.
 2. **Random:** uniformly random sample of 500 nodes; induced subgraph’s largest connected component used (resampled if $|\text{LCC}| < 400$).
- **Location:** Section 5.4 (SNAP Validation), new Table 4.
- **Finding:** Random 500-node sampling fails for both Gnutella datasets — the largest connected component is only 71 nodes (Gnutella08) or ≤ 2 nodes (Gnutella31) even after 20 resampling attempts. Gnutella P2P graphs are ultra-sparse locally (global mean degree < 5), so BFS-hub extraction remains the only viable strategy for $n = 500$ connected subgraphs. Table 4 now documents this limitation explicitly.

Fix 6A — Gossip entropy estimation extended to $k \in \{2, 3, 4, 5\}$

- **Old:** Only $k = 2$ and $k = 3$ hop-neighbourhood sizes studied.
- **New:** Extended to $k \in \{2, 3, 4, 5\}$. New Table 5 reports false-early-trigger rate and missed-trigger rate per $(k, \text{topology})$.
- **Key finding:** Minimum k achieving $< 5\%$ missed-trigger rate differs by topology: $\text{ER} \rightarrow k^* = 2$, $\text{BA} \rightarrow k^* = 3$, $\text{WS} \rightarrow k^* = 4$. Practitioners on WS-like (lattice-heavy) networks should use at least $k = 4$ for reliable distributed triggering.
- **Location:** Section 5.5 (Gossip Estimation), Table 5.

Fix 6B — Symmetric adversarial Laplacian; aggressive threshold

- **Old:** Adversarial Laplacian formed by column-only scaling, breaking graph symmetry ($L \neq L^\top$). Only conservative threshold $H_c = 0.65 H_{\max}$ studied.
- **New (Eq. 13):** Symmetric formulation — for each adversarial node j , *all* incident edges scaled: $A_{ij}^{\text{adv}} = A_{ji}^{\text{adv}} = (1 - f)A_{ij}$. Adversarial–adversarial edges corrected for double-counting; $L^{\text{adv}} = D^{\text{adv}} - A^{\text{adv}}$ remains symmetric PSD.
- **Added:** Aggressive threshold $H_c = 0.50 H_{\max}$ comparison.
- **Results:**

Threshold	BA IEE increase ($f=0.3$)	ER IEE increase ($f=0.3$)
$H_c = 0.65 H_{\max}$	+6.6% (+6.53 units)	+2.1% (+1.26 units)
$H_c = 0.50 H_{\max}$	+4.8% (+3.96 units)	+1.4% (+0.83 units)

- **Interpretation:** Tighter deletion thresholds (more aggressive privacy protection) reduce adversarial leverage because ELAPSE responds earlier, before adversarial nodes can accumulate influence.

Fix 7 — Conclusion bullet contradiction resolved

- **Old:** “ELAPSE . . . outperforms every individual mechanism across all topologies.”
- **Problem:** Directly contradicted by Table 1 (M5 beats M6 on ER, $t = -12.18$, $p < 10^{-13}$) which the paper itself proves.
- **New:** “achieves the lowest topology-averaged IEE across all scales. The M5 mechanism strictly dominates M6 on ER. . . a trade-off discussed in Section 5.4.”

Fix 8 — Train/test caption inconsistency resolved

- **Old caption:** “weights learned on $N_{\text{train}} = 10$ held-out seeds”
- **Actual procedure:** One seed (seed 0) used for weight learning
- **New caption:** “M6 weights learned on training seed 0 only”
- **Reason:** Using all 10 seeds for Nelder-Mead makes each objective evaluation $10\times$ slower; single-seed learning is tractable while still being genuinely held-out from the test set (seeds 10–29).

Fix 9 — Table 4 Gnutella footnote expanded to dedicated subsection

- **Added:** New subsection 6.3 “Topology of Gnutella P2P Networks and Sampling Limitations” explaining why random sampling fails (expected edge count formula, empirical LCC measurements) and clarifying the scientific scope of BFS-hub validation.

Fix 10 — Figure 4 caption corrected

- **Old:** “M6 maintains the largest margin over the best single mechanism at all scales” (false on ER)
- **New:** “M6 achieves the lowest topology-averaged IEE at all scales” (accurate; includes note about ER exception)

Fix 11 — §7 adversarial $n = 200$ disambiguated from §5 $n = 500$

- Added explicit note at the start of Section 7: adversarial experiments use $n = 200$; Table 1 uses $n = 500$; the IEE values are not directly comparable.

Fix 12 — $x_i(0)$ initialisation defined in §3.1

- Added one paragraph to Section 3.1 (Network and Notation) specifying: $x_i(0) \sim \text{Uniform}(0.5, 1.0)$ for 10% randomly chosen source nodes, $x_i(0) = 0$ otherwise; total initial mass $M(0) \in (0, n/10]$; IEE has units [nats $\times$ data-mass $\times$ time].

Fix 13 — References [18] and [20] re-verified and corrected

- **Old [18] / juliussen2023algorithms:** Listed authors Nguyen et al. with DOI 10.1145/3603620. CrossRef shows that DOI belongs to Xu et al. “Machine Unlearning: A Survey” (2023). Corrected to xu2023machine with verified author list.
- **Old [20] / liu2024machine:** Listed authors Liu, Yao, Bai, Papadimitriou with DOI 10.1145/3649464. CrossRef shows that DOI belongs to a spintronic spiking neural networks paper — another fabricated reference. Replaced with nguyen2025survey: Nguyen et al. “A Survey of Machine Unlearning,” ACM TIST 2025, DOI 10.1145/3749987 (verified).

Fix 14 — Table 5 greyscale legibility

- Replaced “Shaded: first k achieving $\text{MS} < 5\%$ ” with a $\checkmark$ column and “ $k^* = 2/3/4$ ” annotations directly in the table, which are legible in B&W print.

Q1 Upgrade — Fix 15: Topology-refined IEE bound (Corollary 2)

- **Added:** Corollary 2 (Topology-refined IEE bound via algebraic connectivity) following Theorem 2.

- **Key result:** The normalised distribution $p(t) = \mathbf{x}(t)/M(t)$ satisfies $\dot{p} = -\alpha Lp$ regardless of mortality (mortality terms cancel). Spectral decomposition + χ^2 -KL inequality gives $H(t) \geq \ln(n) - C_0 e^{-2\alpha\lambda_2 t}$ where $C_0 = n\|p(0) - \mathbf{1}/n\|_2^2$. The spectral trigger time $\tau(\lambda_2) = (2\alpha\lambda_2)^{-1} \ln(C_0/(\ln n - H_c))$ is an explicit upper bound on the trigger time.
- **New bound:** $\text{IEE}(\mathbf{w}) \leq \ln(n)M_0[\tau(\lambda_2) + (1 - e^{-\mu\delta_{\min}(T-\tau)})/(\mu\delta_{\min})]$, a topology-dependent tightening of Theorem 2.
- **Interpretation:** Dense graphs (large λ_2 , small τ) trigger earlier, consistent with empirical $\text{ER} < \text{BA} < \text{WS}$ IEE ordering.

Q1 Upgrade — Fix 16: Residual-mass privacy guarantee (Proposition 3)

- **Added:** Proposition 3 (IEE bound implies residual-mass privacy guarantee) and Remark 5 (relation to differential privacy).
- **Key result:** From $M(T) \leq M_0 e^{-\mu\delta_{\min}T}$, an adversary sampling Q uniform random nodes at time T recovers expected mass fraction $\leq Qe^{-\mu\delta_{\min}T}/n$. Choosing $\mu\delta_{\min} \geq T^{-1} \ln(Qn/\eta)$ ensures recovery probability $\leq \eta/n$ per node — no better than blind random search.
- **Relation to DP:** Formally distinct from differential privacy (DP bounds inference leakage; Proposition 3 bounds physical mass recovery), but shares the exponential-decay parameterisation $\varepsilon(T) = e^{-\mu\delta_{\min}T}$.

Q1 Upgrade — Fix 17: Chronological-timer baseline M7 (Proposition 4)

- **Added:** Proposition 4 (ELAPSE IEE is bounded above by the matched chronological timer) in Section 4.
- **New mechanism:** M7 (Vanish-style fixed-timer deletion) implemented in `src/m7_timer.py`. Timer is calibrated to M6’s mean trigger time $t^*(\mathbf{w}^*)$, giving both baselines identical deletion budgets.
- **Key result:** Since ELAPSE applies gradual mortality from $t = 0$, $H(t; \mathbf{w}^*) \leq H_0(t)$ and $M(t; \mathbf{w}^*) \leq M_0$ throughout $[0, t^*)$, so the pre-trigger IEE integral is strictly smaller than M7’s. The post-trigger residual R is bounded by $\frac{\ln(n)}{2} M_0 \cdot (1 - e^{-\mu\delta_{\min}(T-t^*)})/(\mu\delta_{\min})$, which is dominated by the pre-trigger gap.
- **Subsection 5.6** added: “Comparison with Chronological-Timer Baseline (M7)” in the Numerical Scale Analysis section.

Simulation re-runs. All quantitative results in Sections 5.1–5.5 were re-computed with the corrected code (symmetric Laplacian, train/test split, $k \in \{2, 3, 4, 5\}$, both thresholds). All figures 2–8 were regenerated from the new data. M7 simulation data (Table 6) computed from `m7_timer.py`.